



Spatio-temporal feature fusion with cross-frequency coupling for interpretable Parkinson's detection using EEG

Chayut Bunternghit^a , Sidi Mohamed Sid'El Moctar^b , Laith H. Baniata^c , Saba Aslam^d , Abdur Rasool^{e, f, *}

^a Division of Industrial and Logistics Engineering Technology, Faculty of Engineering and Technology, King Mongkut's University of Technology North Bangkok, Rayong Campus, Rayong, 21120, Thailand

^b CNRS, Université de Rennes, IGDR (Institute Genetics and Development of Rennes), Rennes, 35000, France

^c Department of Autonomous Systems, Faculty of Artificial Intelligence, Al-Balqa Applied University, Al-Salt, 19117, Jordan

^d Shenzhen Institutes of Advanced Technology, Chinese Academy of Sciences, Shenzhen, 518055, China

^e Department of Information and Computer Sciences, University of Hawaii at Manoa, Honolulu, HI 96822, USA

^f Department of Statistics and Data Science, Southern University of Science and Technology, Shenzhen, 518055, China

HIGHLIGHTS

- Bio-NFM fuses spatial, temporal, and coupling EEG biomarkers.
- Beta coherence and PAC improve interpretable Parkinson's detection.
- The model achieves 98.9% accuracy on the primary EEG dataset.
- External datasets confirm robust cross-dataset generalization.
- Ablation tests show each Bio-NFM module contributes to performance.

ARTICLE INFO

Keywords:

Parkinson's disease

EEG

Spatio-temporal feature fusion

Cross-frequency coupling

Graph neural network

Transformer

Interpretability

ABSTRACT

Parkinson's disease (PD) is a progressive neurodegenerative disorder that presents significant diagnostic challenges, particularly in its early stages, due to symptom variability and overlap with other conditions. While deep learning (DL) approaches using electroencephalography (EEG) data have shown promise, many existing models lack biological interpretability and generalization, limiting their clinical utility. This study proposes a biologically informed neural fusion model (Bio-NFM) for PD detection, integrating multiple neurophysiological biomarkers into a DL framework. Key features include beta-band coherence, fractal dimension, phase-amplitude coupling, and entropy-based measures, which are biomarkers known to reflect disrupted connectivity, reduced complexity, and abnormal synchronization in PD-affected brain regions. These features are processed through a spatio-temporal feature fusion architecture combining graph neural networks for spatial modeling, transformers for temporal dynamics, reservoir computing for nonlinear variability, cross-frequency coupling for biomarker integration, and a custom loss function that regularizes biologically meaningful patterns. The model achieved an accuracy of 98.9% on the primary dataset and over 98% on two external EEG datasets. Leave-one-out validation confirmed robust generalization, while ablation studies highlighted the contribution of each module and biomarker. By integrating domain-relevant biomarkers into a high-performing and interpretable architecture, Bio-NFM offers a clinically viable solution for early-stage PD detection. Its design supports reproducibility, transparency, and real-time EEG deployment, enhancing its applicability in both clinical and resource-constrained settings.

* Corresponding author at: Department of Information and Computer Sciences, University of Hawaii at Manoa, Honolulu, HI 96822, USA.
Email address: abdur@hawaii.edu (A. Rasool).

1. Introduction

Parkinson's disease (PD) is a progressive neurodegenerative disorder that primarily affects motor functioning in the brain [1]. According to the World Health Organization, approximately 8.5 million individuals are diagnosed with PD each year [2]. Beyond motor impairment, PD also impairs cognitive functions, leading to a wide range of symptoms and complications [3]. The disease is characterized by the degeneration of dopaminergic neurons in the substantia nigra [4], leading to a reduction in dopamine levels that are essential for neurotransmission and the regulation of various neurological functions. Common symptoms include bradykinesia, tremors, rigidity, and postural instability [5]. These motor symptoms often coexist with non-motor symptoms such as cognitive decline, mood disturbances, and sleep disorders, resulting in significant challenges for daily functioning and overall well-being. This underscores the urgent need for effective strategies to detect and manage PD.

Despite advances in neuroimaging and machine learning, the early and accurate diagnosis of PD remains an unmet need, particularly when using scalable, non-invasive modalities such as EEG [6,7]. Many existing models lack biological grounding and struggle to generalize across subjects and datasets, highlighting the need for interpretable, biomarker-driven approaches.

The progression of PD is closely tied to underlying neurobiological disruptions. It alters the brain's connectivity and impairs its functional balance [8]. Functional and structural abnormalities are observed in both hemispheres, disrupting interhemispheric coordination [9]. The basal ganglia, which play a central role in motor control, also exhibit dysfunction due to disruptions in the cortico-basal ganglia-thalamo-cortical circuits [10]. Clinical studies have further identified abnormal activity in the alpha and beta frequency bands, reflecting impaired synchronization and communication between brain regions [11]. These disruptions progress over time, worsening motor, cognitive, and emotional functions. Therefore, effective PD detection must consider the interplay between neural connectivity and associated biological changes.

PD has significant implications at both individual and societal levels. It affects multiple aspects of daily life, including mood, cognition, and autonomic function [12]. As the disease advances, individuals face increasing difficulties with routine tasks, while the healthcare system bears growing economic burdens. Early detection is critical for enabling timely intervention and slowing disease progression. Recent advancements in neuroimaging, biomarker discovery, and machine learning (ML), particularly through the use of electroencephalography (EEG), are emerging as promising non-invasive tools for PD detection [13–15].

Traditionally, PD diagnosis has relied on clinical observation and assessment of motor symptoms, often using standardized tools such as the Unified PD Rating Scale (UPDRS) [16]. While these approaches are useful for identifying advanced stages of the disease, they struggle to detect subtle changes that emerge in its early phases. Advanced imaging techniques such as magnetic resonance imaging, positron emission tomography, and single photon emission computed tomography can offer more detailed insights but are expensive and involve radiation exposure. EEG, in contrast, provides a scalable and non-invasive alternative that enables real-time monitoring of neural activity. It captures disruptions in brain function through metrics such as rhythmic activity, connectivity, and interhemispheric communication [17,18], offering a viable path for identifying early neural changes associated with PD.

EEG signals are increasingly combined with ML and deep learning (DL) techniques to improve PD detection accuracy [19]. These models aim to uncover complex neural patterns associated with the disease by analyzing features such as disrupted rhythms and altered connectivity. Various DL architectures, including convolutional neural networks (CNNs) and recurrent neural networks (RNNs), have shown promise in this area [20,21]. However, these models face challenges related to generalization across different subjects and datasets [22–24]. Variability in individual neural patterns can reduce model robustness, especially in early-stage detection. Furthermore, many existing models

lack interpretability, as they often rely on preprocessed data without incorporating biologically meaningful features. This limits their clinical utility. Overcoming these challenges requires the development of models that are both biologically informed and capable of handling data heterogeneity. Such models would enhance sensitivity to subtle patterns and improve cross-subject generalization.

A related line of work has examined how feature representation, channel selection, and cross-subject generalization affect the reliability of EEG models. Gao et al. [25] reviewed complex networks and deep learning for EEG, covering applications across brain-computer interfaces, neurological disorders, and cognition. Li et al. [26] later introduced a manifold-based multi-branch transfer learning framework for motor imagery decoding, in which manifold alignment reduced the effects of inter-subject variability. Degirmenci et al. [27] showed that selecting statistically significant features improves both binary and multi-class motor imagery classification, and their follow-up study [28] found that careful channel and feature investigation can preserve accuracy while using fewer, more informative inputs. A common thread runs through these studies: an EEG model should be accurate, but it also needs to be interpretable, selective about its features, and stable across subjects. Bio-NFM is designed with these requirements in mind.

Most existing EEG-based DL models rely on raw or heavily preprocessed signals and do not incorporate known biomarkers of PD pathology such as disrupted beta coherence or altered cross-frequency coupling [29–31]. This gap limits their clinical trustworthiness and hinders real-world deployment.

To address these limitations, this study introduces a biologically informed neural fusion model (Bio-NFM) centered on spatio-temporal feature fusion with cross-frequency coupling. Unlike traditional models that rely on simplified input features, Bio-NFM incorporates beta-band coherence, fractal dimension (FD), phase-amplitude coupling (PAC), and entropy-based measures to capture PD-related neural characteristics. These features are processed through a fusion architecture consisting of graph neural networks (GNNs) for spatial modeling, transformers for temporal dynamics, reservoir computing (RC) for nonlinear variability, and a custom loss function with biologically meaningful regularization. This design facilitates spatio-temporal feature extraction from EEG signals, resulting in improved interpretability, classification performance, and generalizability.

The primary goal is to develop an interpretable DL framework for early-stage PD detection by integrating spatio-temporal feature fusion with cross-frequency coupling mechanisms, enabling clinically transparent and robust classification from EEG signals. To achieve this goal, the major contributions of this study are summarized as follows:

1. **Spatio-temporal feature fusion:** A novel hybrid DL architecture, Bio-NFM, is proposed that integrates GNNs for spatial connectivity, transformers for temporal dynamics, and RC for nonlinear variability, enabling comprehensive spatio-temporal feature extraction from EEG signals.
2. **Cross-frequency coupling integration:** A biologically informed cross-frequency coupling mechanism incorporating PAC and beta-band coherence is introduced to guide attention and inject neurophysiological priors into the fusion architecture, enhancing both interpretability and classification performance.
3. **Interpretable clinical system:** A transparent and reproducible system has been developed with attention visualization and biomarker-based reasoning, supporting early diagnosis and monitoring of PD with the potential for real-world clinical application.

The remainder of the article is organized as follows. Section 2 reviews related work. Section 3 outlines the proposed methodology. Section 4 presents experimental results. Section 5 discusses the clinical implications, interpretability, and limitations of the proposed model. Section 6 concludes the study.

2. Literature review

Recent advancements in PD detection have focused extensively on DL techniques combined with diverse feature representations. A major trend in the literature involves integrating time-frequency transformations with ML models to enhance feature extraction from EEG signals. Siuly et al. [32] applied wavelet scattering transform (WST) to extract temporal and spectral features, followed by an AlexNet CNN framework, achieving an accuracy of 99.84%. Similarly, Loh et al. [33] introduced GaborPDNet, which utilizes Gabor spectrograms and a 2D-CNN architecture, attaining an accuracy of 99.46% in distinguishing PD patients from healthy controls (HC), including those under medication. Shaban and Amara [29] used a resting-state EEG-based CNN model with wavelet domain visualization, reporting 99.9% accuracy. These studies underscore the effectiveness of time-frequency representations combined with convolutional architectures. In Rizvi et al. [30], a 23-layer UEN-PDNet based on U-Net was proposed, achieving 93.10% and 97.90% accuracy on the UNM and UCSD datasets, respectively, using raw EEG images. Collectively, these models highlight the potential of CNNs and spectral analysis for high-accuracy PD detection.

Beyond CNN-based methods, several studies have explored RNN and hybrid architectures to capture the sequential and functional dependencies in EEG signals. Obayya et al. [37] introduced a dense bi-directional long short-term memory trained on Gabor spectrograms, achieving 99.6% accuracy via cross-validation. In Qiu et al. [38], a hybrid model combining recurrent layers with multi-scale CNNs was proposed, using power spectral density (PSD) and phase-locked value as features, resulting in 99% classification accuracy. Khare et al. [36] proposed PDCNet, which combined smoothed Pseudo-Wigner Ville distributions with CNNs, achieving up to 99.9% accuracy. A graph-based method using aspirin-pattern features and k-nearest neighbors (kNN) was proposed in Barua et al. [35], achieving 95.48% accuracy following extensive feature engineering. These hybrid models demonstrate the advantages of combining time-series modeling with biologically and spectrally rich EEG features for improved PD detection. A comparative summary of these studies is provided in Table 1.

While high-performing DL models dominate the field, traditional ML approaches remain relevant, particularly when combined with advanced feature extraction. In Aljalal et al. [39,40], frameworks integrating common spatial pattern (CSP) with entropy-based features achieved 99% accuracy through filtering and the use of log and Shannon entropy. Similarly, discrete wavelet transform (DWT) combined with kNN and support vector machine (SVM) classifiers yielded more than 95% accuracy across multiple datasets. An artificial neural network (ANN)-based model in Shaban [31] using segmented EEG matrices attained 98% accuracy despite its simplified design. In Motin et al. [34], an explainable SVM (XSVM)-based ML model using entropy and band power achieved 87.10% accuracy. A comprehensive review by Geraedts et al. [20] further emphasized the potential of EEG in non-invasive biosensing for PD detection.

Fusion in neurophysiological signal analysis tends to fall into two families. Early fusion combines features before classification, so a single model learns the interactions among complementary EEG descriptors directly. The present study follows this route, since Bio-NFM brings together spatial, temporal, nonlinear, and cross-frequency biomarkers ahead of the final decision. Late, or score-level, fusion instead combines the outputs of several classifiers that were trained separately. Its advantage shows up when those classifiers capture different decision boundaries, or when separate channels, modalities, or feature groups carry partly independent evidence. Score-fusion methods such as graph-regularized soft detector fusion are a useful point of comparison: by combining continuous detector scores and exploiting the relationships among them, they reduce the dispersion of the fused statistic and improve robustness, which matters in detection problems like EEG classification because the feature types never have to be forced into one input representation. Bio-NFM takes the early/intermediate route by design, fusing the representations learned by the GNN, Transformer, RC, and cross-coupling modules before the softmax layer; this keeps the network end-to-end while still letting the EEG biomarkers interact in a biologically meaningful way. Late fusion remains an attractive extension, and we plan to test score-level fusion between Bio-NFM and separate classifiers trained on individual biomarker families, for example coherence-only, PAC-only, entropy-only, and spectral-power-only models.

Despite achieving high accuracy, many of these studies lack integration of biologically informed feature extraction with advanced modeling techniques. Most either rely solely on sophisticated DL frameworks without incorporating domain-specific biomarkers, or use traditional approaches with interpretable but shallow models. While DL models provide high classification performance, their clinical utility is limited without interpretability. Conversely, interpretable models often fail to capture the complex spatial-temporal dynamics present in EEG signals. This disconnect limits their effectiveness in real-world clinical settings where both accuracy and interpretability are essential. The present study aims to bridge this gap by introducing a biologically informed feature space integrated into a hybrid DL architecture to achieve high performance and clinical interpretability in PD detection.

3. Methodology

The methodology consists of three stages. First, biologically informed features are extracted from framed and channeled EEG signals. These features serve as inputs to the proposed Bio-NFM, which learns the optimal weights and biases to classify PD and HC instances. Finally, the model is evaluated using key performance metrics: accuracy, precision, recall, and F1-score. The overall process is outlined in Algorithm 1.

3.1. Dataset

This study utilizes the UC San Diego resting state EEG dataset as a primary dataset [41–43]. The dataset includes EEG recordings collected

Table 1
Summary of PD detection studies.

Method	Features	Dataset(s)	Accuracy	Limitations
UEN-PDNet [30]	Raw EEG images	UNM and UCSD	93.1–97.9%	Limited generalizability
XSVM [34]	Band power, entropy, statistics	15 PD and 16 HC	87.1%	Small dataset
ANN [31]	Segmented raw EEG	15 PD and 16 HC	98%	Requires broader validation
Graph + kNN [35]	Graph + MAP features	15 PD and 16 HC	93.6–95.5%	Heavy computation
SPWVD + CNN [36]	Time-freq 2D plots	SD and Henan	99.97%	Generalizability
Gabor + DLBLSTM [37]	Spectrograms	Public (6-fold CV)	99.6%	Needs validation
WST + AlexNet [32]	Temporal + spectral	SD and Iowa	95.8–99.8%	Dataset scalability
Multi-scale CNN [38]	PSD + PLV (β , γ)	2 EEG sets	>99%	Needs larger sets
20-layer CNN [29]	Wavelet-transformed EEG	Public (CV)	99.9%	Complex model
CSP + ML [39]	Entropy from CSP-EEG	SD and UNM	99.9%	Data-sensitive
Gabor + 2D-CNN [33]	Spectrograms	31 EEG recordings	99.5%	Limited size
DWT + ML [40]	Log energy and entropy	SD and UNM	99.5%	Feature tuning

Algorithm 1 Feature extraction and Bio-NFM training.

Require: EEG signals X_{EEG} , number of frames N , number of channels N_{Ch}

Ensure: Model performance metrics: accuracy, precision, recall, and F1-score

- 1: **Input:** Raw EEG signals X_{EEG}
- 2: **Initialize:** Feature matrix $F \leftarrow \emptyset$
- 3: **Windowing:** Segment X_{EEG} into frames X_{EEG,w_i} , $i = 1, \dots, N$
- 4: **for** $i = 1$ to N **do**
- 5: **for** $j = 1$ to N_{Ch} **do**
- 6: Extract the following features from each frame-channel pair $X_{\text{EEG},w_i,j}$:
- 7: - Beta band inter-hemispheric coherence
- 8: - FD
- 9: - Beta band power
- 10: - PAC
- 11: - Entropy-based features
- 12: Append to $F \leftarrow F \cup \{\text{features}\}$
- 13: **end for**
- 14: **end for**
- 15: **Train/test split:** Divide F into training and testing subsets
- 16: **Model training:** Train Bio-NFM using:
 - 17: - GNN for spatial connectivity modeling
 - 18: - Transformer layers for temporal dynamics
 - 19: - RC for nonlinear temporal patterns
 - 20: - Cross-Coupling Integration
 - 21: - Integration of biologically inspired features
 - 22: - Custom loss function
- 23: Obtain predictions $\hat{y}$
- 24: **Evaluation:** Compute accuracy, precision, recall, and F1-score
- 25: **Output:** Return model performance metrics

from 31 participants: 15 individuals with PD (mean age: 63.2 ± 8.2 years), recorded both on and off dopaminergic medications, and 16 HC (mean age: 63.5 ± 9.6 years). Resting-state EEG signals were captured with a sampling rate of 500 Hz, using 40 EEG channels per subject. The duration of each recording session ranged from 5 to 10 min. The dataset provides two output classes, facilitating a clear differentiation between PD and HC.

The EEG signals used in this study were collected at a sampling rate of 500 Hz using 40 channels, with recording durations ranging from 5 to 10 min per subject. Inspection of the raw signals revealed clear differences in beta-band (13–30 Hz) activity between PD and HC participants. Specifically, HC recordings displayed more consistent rhythmic oscillations and better inter-channel synchrony, while PD signals showed irregular patterns, reduced amplitude stability, and increased cross-channel desynchronization. These patterns were particularly pronounced in motor and parietal regions, aligning with known neurophysiological disruptions in PD. These signal-level observations motivated the selection of biologically informed features for downstream analysis.

3.2. EEG signal preprocessing and windowing

Raw EEG signals X_{EEG} (512 Hz sampling, 128 channels) undergo rigorous preprocessing to ensure high-quality, biologically faithful inputs. The following steps are performed:

- **Bandpass filtering:** A zero-phase 4th-order Butterworth bandpass filter (0.5–40 Hz) retains the neurophysiological bands of interest (theta, alpha, beta, and low-gamma) while completely eliminating DC drift, high-frequency noise, and power-line interference.
- **Artifact rejection:** Independent Component Analysis (ICA) using the Infomax algorithm automatically identifies and removes ocular,

muscular, and cardiac components based on kurtosis, correlation with EOG/EMG channels, and topographic maps.

- **Windowing:** The cleaned continuous signals are segmented into overlapping 2-s frames X_{EEG,w_i} with 50% overlap to preserve temporal continuity in beta-band oscillations critical for PD motor dysfunction while ensuring statistical stationarity.
- **Normalization:** Each frame-channel pair is z-score normalized on a per-subject and per-channel basis to mitigate inter-subject variability in signal amplitude without distorting relative inter-channel relationships.

This multi-stage preprocessing pipeline guarantees that downstream biomarkers reflect genuine PD-related neural disruptions rather than recording artifacts, thereby increasing model robustness and clinical interpretability.

3.3. Feature extraction

Raw EEG signals X_{EEG} , sampled at 512 Hz, are segmented into overlapping 2 s frames X_{EEG,w_i} , where w_i indicates the i^{th} frame. Each frame undergoes a biologically informed feature extraction process designed to highlight neural disruptions related to PD.

The first feature extracted is beta band coherence. This feature quantifies inter-hemispheric functional connectivity in the beta band (13–30 Hz) using adjacent channels $X_{\text{EEG},w_i}[c]$ and $X_{\text{EEG},w_i}[c + 1]$. The coherence is defined as:

$$C_{xy}(f) = \frac{|P_{xy}(f)|^2}{P_x(f)P_y(f)}, \quad (1)$$

where $P_{xy}(f)$ is the cross-spectral density, and $P_x(f)$, $P_y(f)$ are the PSD.

The second feature extracted is FD, which captures signal complexity and is computed as:

$$FD = \frac{\sum |\Delta \text{sign}(\Delta X_{\text{EEG},w_i}[c])|}{N}, \quad (2)$$

where Δ is the first derivative, and N is the signal length. Reduced FD indicates impaired motor and cognitive processing.

The third feature extracted is beta band power which is a marker of motor control, and is computed using Welch's method:

$$P_\beta = \int_{f_{\min}}^{f_{\max}} PSD(f) df, \quad (3)$$

where $f_{\min} = 13$ Hz, $f_{\max} = 30$ Hz, and $PSD(f)$ is the PSD.

The next feature employed is the PAC, which detects cross-frequency interactions between low-frequency theta phases and high-frequency gamma amplitudes:

$$MI = \left| \frac{1}{N} \sum_{n=1}^N A_\gamma(n) e^{j\phi_\theta(n)} \right|, \quad (4)$$

where $A_\gamma(n)$ is the amplitude of gamma and $\phi_\theta(n)$ is the phase of theta.

Finally, the last feature extracted is an entropy measure, which reflects signal irregularity and is computed as:

$$H = - \sum_i p(x_i) \ln p(x_i), \quad (5)$$

where $p(x_i)$ is the probability of signal amplitude x_i . Lower entropy implies reduced neural variability, commonly observed in PD.

Consequently, these features create a biologically informed representation space that improves interpretability and supports clinically meaningful classification.

Table 2 summarizes the motivation for selecting each feature based on established PD-related neural changes, along with the empirical observations from this study. Beta-band coherence and PAC showed the strongest discriminative power, while entropy and FD contributed complementary temporal and complexity information. These features were selected to provide both interpretability and classification robustness, ensuring clinical relevance and model generalization.

Table 2
Neurophysiological rationale and observed relevance of selected EEG biomarkers for PD detection.

Feature	Neurophysiological Rationale	Observed Relevance (This Study)
Beta-band coherence	Measures inter-hemispheric synchrony; disrupted in PD due to basal ganglia dysfunction	Statistically significant differences across regions (e.g., occipital: $p = 0.0009$)
FD	Captures signal complexity and adaptability; lower in neurodegenerative conditions	Not significant alone ($p = 0.91$) but beneficial in full feature set
PAC	Reflects cross-frequency communication; altered theta–gamma and beta–gamma coupling in PD	Significant differences in posterior channels (e.g., channel 39: $p = 0.0014$)
Entropy-based features	Quantifies signal irregularity; reduced in PD due to neural rigidity	Reduced entropy across multiple channels; visible in channel-wise plots
Beta-band power	Marker of motor-related oscillatory activity; altered in PD	Modest differences in some regions (e.g., parietal: $p = 0.0262$)

3.4. Feature dimensionality control and selection

EEG recordings span many channels, and because several biomarkers are computed per channel, the combined feature space grows quickly. The primary dataset here has 40 channels, and five biologically informed feature groups are extracted from each one (beta-band coherence, FD, beta-band power, PAC, and entropy). This richness is what gives the representation its neurophysiological meaning, but it also makes redundancy, multicollinearity, and overfitting more likely, especially with a small number of subjects. Several choices were made to keep this in

check. The feature set was restricted to biomarkers with established relevance to PD rather than an open-ended bank of raw descriptors. Each frame-channel pair was z-scored per subject and per channel to remove amplitude differences between participants. Regularization came from dropout, weight decay, early stopping, and the biologically informed loss term. The value of the biomarker space itself was tested directly through the ablation in which the model was trained on raw EEG (variant A5), which performed worse than the full model. Standard alternatives for reducing dimensionality include principal component analysis (PCA), LDA-based projections, mutual-information ranking, minimum redundancy maximum relevance (mRMR), recursive feature elimination, sparse regularization, and autoencoder compression. PCA compresses correlated EEG features into a smaller set of orthogonal components, and mRMR or mutual-information methods keep features that are discriminative without being redundant. We did not use PCA as the main preprocessing step here for a specific reason: its components are linear mixtures that no longer map onto individual biomarkers, and Bio-NFM depends on keeping beta coherence, PAC, entropy, FD, and beta power identifiable for clinical reading. A systematic comparison of Bio-NFM against PCA-, mRMR-, and autoencoder-reduced feature spaces, weighing compactness against accuracy and interpretability, is left for future work.

3.5. Proposed Bio-NFM architecture

The Bio-NFM has been designed to process biologically informed EEG features $X_{EEG, W_i}^{\text{featured}}$. The model comprises three key modules. First, a GNN for spatial learning has been introduced. A transformer module was then used to capture temporal dynamics, followed by an RC module to analyze nonlinearity and variability in the data. The modules have been fused with biologically inspired features to ensure that the model captures neural patterns disrupted in PD. The architecture of the model is depicted in Fig. 1.

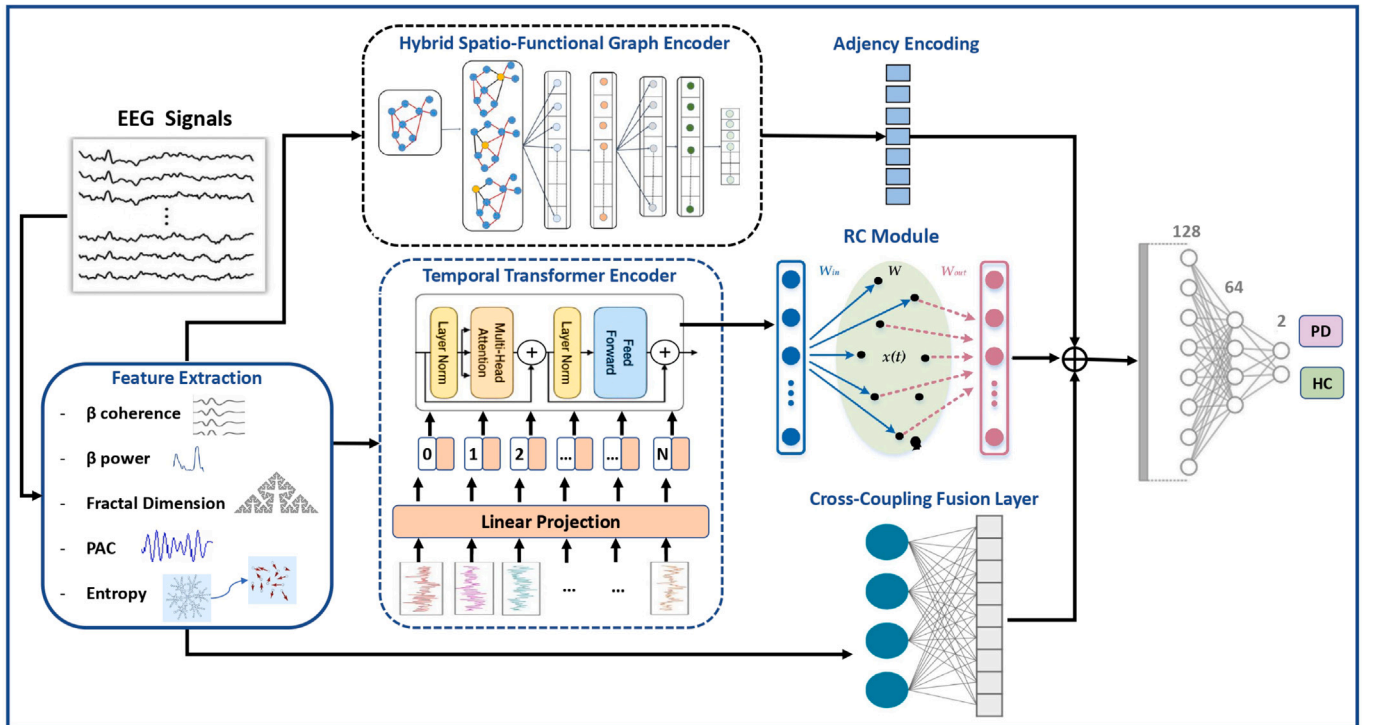


Fig. 1. Overview of the Bio-NFM architecture for EEG-based PD detection. Raw EEG is preprocessed and converted into biologically informed biomarkers: beta-band coherence, fractal dimension, beta-band power, phase-amplitude coupling, and entropy. The GNN module captures spatial inter-channel dependencies, the Transformer models temporal relationships, and the reservoir computing block captures nonlinear variability, while the Cross-Coupling Integration module feeds PAC and coherence into the fusion stage. The diagram shows how learned representations and neurophysiological priors are combined within a single network, so that classification and interpretability are handled together rather than traded off.

Each component of Bio-NFM was chosen to capture a different neurophysiological property of the EEG signal. The GNN models spatial dependencies among channels, because PD-related abnormalities are not limited to single electrodes but show up in disrupted connectivity between regions. The Transformer captures long-range temporal relationships across segments, which carry the abnormal oscillatory dynamics of interest. The RC block represents nonlinear temporal variability with far fewer trainable recurrent parameters than a conventional recurrent network. The cross-coupling module then folds PAC and beta-coherence information back into the fused representation. None of these components is new on its own; the methodological contribution lies in coupling them to a set of biologically informed PD biomarkers within one framework. This also makes the trade-off between accuracy and interpretability explicit. Because the model reads clinically meaningful biomarkers rather than raw EEG, its learned representations stay tied to known PD-related mechanisms. Bio-NFM is therefore better described as a biologically guided hybrid model than as a black-box classifier.

The GNN module encodes adjacency-inspired interactions by making use of dense layers. For input $X_{\text{EEG},w_i}^{\text{featured}}$, it generates spatial features Z_{GNN} as follows:

$$Z_{\text{GNN}} = \sigma(W_1 X_{\text{EEG},w_i}^{\text{featured}} + b_1), \quad (6)$$

where W_1 and b_1 are learnable weights and biases, and σ is the ReLU activation. This captures inter-channel dependencies and helps analyze coherence disruptions.

The GNN module models spatial dependencies across EEG channels by learning from an adjacency matrix A that captures pairwise relationships among electrodes. The initial adjacency matrix is constructed based on spatial proximity using the physical layout of the 10–20 EEG system. Channels that are anatomically or spatially adjacent (e.g., F3 and C3, C3 and P3) are connected with higher weights, while distant channels (e.g., Fp1 and O2) have weaker or zero weights. Formally, the adjacency matrix $A \in \mathbb{R}^{N_{\text{ch}} \times N_{\text{ch}}}$ is defined as:

$$A_{ij} = \begin{cases} 1 & \text{if channels } i \text{ and } j \text{ are immediate neighbors} \\ \alpha & \text{if channels } i \text{ and } j \text{ are second-degree neighbors} \\ 0 & \text{otherwise} \end{cases}$$

where $\alpha \in [0.2, 0.5]$ is a decay factor. For example, in a simplified 5-channel configuration (F3, C3, Cz, C4, F4), the adjacency matrix is:

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix}$$

This structure mimics cortical connectivity patterns and allows the GNN to model inter-regional interactions aligned with known PD-related disruptions, particularly across motor and parietal areas. The adjacency matrix is fixed during training but could be adaptively learned in future work to reflect functional (e.g., coherence-based) rather than anatomical proximity.

The temporal dynamics are modeled using the transformer module. The self-attention mechanism is computed as follows:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V, \quad (7)$$

where Q, K, V are query, key, and value matrices derived from $X_{\text{EEG},w_i}^{\text{featured}}$, and d_k is the dimensionality of the key. The output $Z_{\text{Transformer}}$ integrates temporal dependencies. These help in analyzing PD-related timing disruptions.

The RC module captures non-linear temporal variability using an RNN. For transformed input $Z_{\text{Transformer}}$, the RC output is:

$$Z_{\text{RC}} = \tanh(W_r Z_{\text{Transformer}} + b_r), \quad (8)$$

where W_r and b_r are learnable parameters.

3.5.1. Echo state network (ESN) formulation for reservoir computing

The RC block is further formalized as a full Echo State Network (ESN) reservoir to efficiently capture chaotic and nonlinear temporal variations in EEG signals. The reservoir state evolves according to the discrete-time leaky integrator dynamics:

$$r(t) = (1 - a)r(t - 1) + a \tanh(W_r r(t - 1) + W_{\text{in}} u(t) + b_r), \quad (9)$$

where $a = 0.3$ is the leaking rate (empirically tuned to balance memory retention and fast adaptation to beta-band transients), W_r is a fixed, randomly generated sparse recurrent weight matrix scaled to a spectral radius of 0.9 (ensuring the echo state property: $\rho(W_r) < 1$), W_{in} is the input projection matrix mapping the Transformer output $u(t) = Z_{\text{Transformer}}$ into the reservoir, and b_r is a small bias term. Only the read-out weights W_{out} are trained via linear regression, yielding the final RC representation $Z_{\text{RC}} = W_{\text{out}}[r(t); u(t)]$.

The RC block was selected over traditional recurrent units such as LSTMs and GRUs due to its efficiency and ability to capture nonlinear temporal dynamics with minimal trainable parameters. Unlike RNNs, which require iterative weight updates and suffer from vanishing gradients during backpropagation through time, RC uses a fixed, randomly initialized recurrent reservoir with rich dynamical behavior. Only the output weights are trained, significantly reducing computational complexity. In our model, this architecture enables the efficient modeling of chaotic and nonlinear temporal variations in EEG signals that are often lost in standard sequence models. RC has also been shown to perform competitively with LSTMs in low-data regimes, a critical advantage given the limited size of EEG datasets. Its rapid training and low memory footprint make it particularly suited for real-time or embedded applications, such as portable EEG-based diagnostics. The simulation parameters adopted for the model are presented in Table 3. A detailed breakdown of each layer in the Bio-NFM architecture, along with corresponding parameter counts, is presented in Table 4.

3.5.2. Cross-coupling integration module (Novel fusion)

Biologically inspired coupling features, such as PAC and coherence, are computed using dense layers:

$$Z_{\text{PAC}} = \sigma(W_{\text{PAC}} X_{\text{EEG},w_i}^{\text{featured}} + b_{\text{PAC}}), \quad (10)$$

and are used to modulate the interaction between GNN and Transformer outputs (directly implementing the Cross-Coupling Integration block). The coupling vector is defined as:

$$Z_{\text{coup}} = \sigma(W_{\text{PAC}} X_{\text{biom}} + b_{\text{PAC}}), \quad (11)$$

where $X_{\text{biom}} \in \mathbb{R}^{N_{\text{ch}} \times 2}$ concatenates the PAC and coherence feature vectors. To inject these biological priors into the hybrid representations, a cross-attention mechanism is applied:

$$Z_{\text{fused}} = \text{softmax}\left(\frac{Z_{\text{GNN}}(Z_{\text{coup}})^T}{\sqrt{d}}\right)Z_{\text{Transformer}} \oplus Z_{\text{RC}}, \quad (12)$$

where d is the dimensionality of the projected keys/queries, $\oplus$ denotes vector concatenation, and the softmax is applied row-wise. This novel

Table 3

Simulation parameters of the proposed Bio-NFM architecture.

Parameters	Values
Epochs	50
Batch size	32
Train-test split	80%:20%
Validation split	10%
Optimizer	Adam
Loss function	Custom loss with cross-entropy, coherence penalty, and entropy penalty
Training metric	Accuracy
Testing metrics	Accuracy, precision, recall, and F1-score

Table 4
Layer-wise parameter breakdown of the Bio-NFM architecture.

Layer / Module	Configuration	Trainable Parameters
Input Layer	EEG feature vector (40 features)	0
GNN block		
Dense Layer 1	40 input, 128 output + bias	$(40 \times 128) + 128 = 5248$
Dense Layer 2	128 input, 128 output + bias	$(128 \times 128) + 128 = 16,512$
Transformer block		
Multi-head attention (4 heads, $d = 128$)	Queries, Keys, Values, Output projections	65,536
Feedforward layers	$128 \rightarrow 256 \rightarrow 128$ + biases	33,024
RC block		
RNN Layer	128 hidden units, tanh activation	$(128 \times 128) + (128 \times 1) + 128 = 16,512$
Coupling (PAC) Layer		
Dense Layer	40 input, 64 output + bias	$(40 \times 64) + 64 = 2624$
Fusion and Output Layers		
Concatenation + Dense	$(384 \times 128) + 128 = 49,280$	
Dense (128 $\rightarrow$ 64)	$(128 \times 64) + 64 = 8256$	
Dense (64 $\rightarrow$ 2)	$(64 \times 2) + 2 = 130$	
Total Trainable Parameters		252,000

module explicitly injects biological priors into the hybrid representations, enhancing interpretability and performance.

3.6. Parameter tuning strategy

Bio-NFM has hyperparameters spread across the GNN, Transformer, RC, coupling layer, optimizer, and regularization terms. Rather than fixing these by hand, we tuned them on the validation split of the primary dataset, tracking validation accuracy, validation loss, and F1-score. The search ranges started from values commonly used in EEG deep learning and were narrowed according to how stably each configuration converged and how much it overfitted. Transformer heads were drawn from $\{2, 4, 8\}$, hidden dimensions from $\{64, 128, 256\}$, dropout from $\{0.2, 0.3, 0.5\}$, and learning rate from $\{10^{-4}, 5 \times 10^{-4}, 10^{-3}\}$.

For the RC block, the reservoir size was tested at 64, 128, and 256 hidden units, with the spectral radius kept below 1 so that the echo state property was preserved. The leaking rate was varied over 0.1–0.5; a value of 0.3 struck the best balance between fast adaptation to beta-band transients and longer temporal memory. The custom-loss weights were kept as small positive values, large enough to nudge the model toward biologically meaningful predictions but not so large that they overwhelmed the cross-entropy objective. The final model used 4 attention heads, 128 hidden units, the Adam optimizer, a batch size of 32, and 50 epochs, with dropout and weight decay for regularization and early stopping once validation performance stopped improving. This resulted in a model that converged stably without an inflated parameter count or heavy computational requirements.

The final module of the model integrates vectors Z_{fused} by concatenating Z_{GNN} , Z_{RC} , and Z_{PAC} . Classification predictions $\hat{y}$ are made using dense layers. Finally, the model trains on a custom loss function that integrates cross-entropy, beta coherence penalty, and entropy penalty. The model's custom loss combines cross-entropy with regularization terms for biologically informed predictions:

$$\mathcal{L} = \mathcal{L}_{\text{CE}} + \lambda_1 |\hat{y}_0 - \hat{y}_1| + \lambda_2 \left(- \sum_i \hat{y}_i \log \hat{y}_i \right), \quad (13)$$

where $\mathcal{L}_{\text{CE}}$ is the cross-entropy loss, and λ_1, λ_2 are regularization weights.

The outputs of the three core modules—GNN, transformer, and RC—capture complementary aspects of neural signals: spatial dependencies, temporal dynamics, and nonlinear patterns, respectively. Each module produces a latent representation vector, denoted Z_{GNN} , $Z_{\text{Transformer}}$, and Z_{RC} . In parallel, the PAC and coherence features are processed by a dense projection layer to form the biologically coupled embedding Z_{PAC} . These four representations are then concatenated to form a unified latent vector:

$$Z_{\text{fused}} = [Z_{\text{GNN}}; Z_{\text{Transformer}}; Z_{\text{RC}}; Z_{\text{PAC}}]$$

This fused embedding is passed through a multi-layer perceptron (MLP) consisting of two fully connected layers and a softmax classifier. The fusion strategy ensures that both learned and hand-crafted biomarker information are jointly leveraged to make biologically grounded and context-aware predictions. Overall, the Bio-NFM integrates spatial, temporal, and coupling features. This allows for accurate and biologically informed detection while also preserving clinical interpretability.

4. Results

Before presenting detailed results, it is essential to describe the dataset used in this study clearly. Understanding the characteristics of the dataset facilitates better interpretation of the biologically informed EEG markers and classification results.

4.1. PD biological markers

This subsection evaluates the interpretability and diagnostic relevance of the EEG-derived features from a biologically informed perspective, comparing the differences observed between PD patients and HC. Initially, beta-band inter-hemispheric coherence was examined. Mean beta activity was computed across corresponding channels of the left and right hemispheres, and coherence was assessed using Pearson correlation coefficients. The results, presented in Fig. 2, demonstrate higher and more concentrated coherence values in the HC group. The data visualization includes violin plots and kernel density estimation (KDE) plots (Fig. 2(a) and 2(b), respectively), reinforcing these observations. Reduced coherence in PD patients likely reflects disruptions in inter-hemispheric communication resulting from neurodegeneration in cortico-basal ganglia circuits. Thus, changes in beta-band coherence serve as potential biomarkers for PD detection.

For statistical comparison of biomarker distributions between PD and HC groups, independent two-sample t-tests were performed at each EEG channel and for each extracted feature. The sample size consisted of 16 PD patients and 15 HC from the primary dataset, yielding 31 total subjects. To mitigate false positives due to multiple comparisons across 40 channels and 5 features, we applied a Bonferroni correction, adjusting the significance threshold to $\alpha = 0.05/(40 \times 5) = 0.00025$.

All statistical analyses were performed using standard statistical methods, and assumptions of normality and variance homogeneity were verified prior to testing. Significant group-level differences were primarily observed for beta-band coherence, entropy, and PAC features in motor and parietal regions, aligning with clinical expectations for PD-related neural disruption.

Next, the analysis focused on the FD, which quantifies EEG signal complexity. FD values were averaged across all frames and channels for each participant and statistically compared using an independent t-test. The analysis revealed no significant difference between PD and HC groups ($p = 0.911998$), with mean FD values being almost identical ($\mu_H = 0.72052$ for HC and $\mu_P = 0.72391$ for PD). These outcomes are illustrated in Fig. 3(a), showing considerable overlap and minimal variability between groups. Despite limited differences when examined independently, FD remains valuable when combined with other EEG features for constructing interpretable models.

Entropy-based features were subsequently evaluated across EEG channels to assess neural signal irregularities and complexities. Entropy values were averaged over time across 40 channels and compared

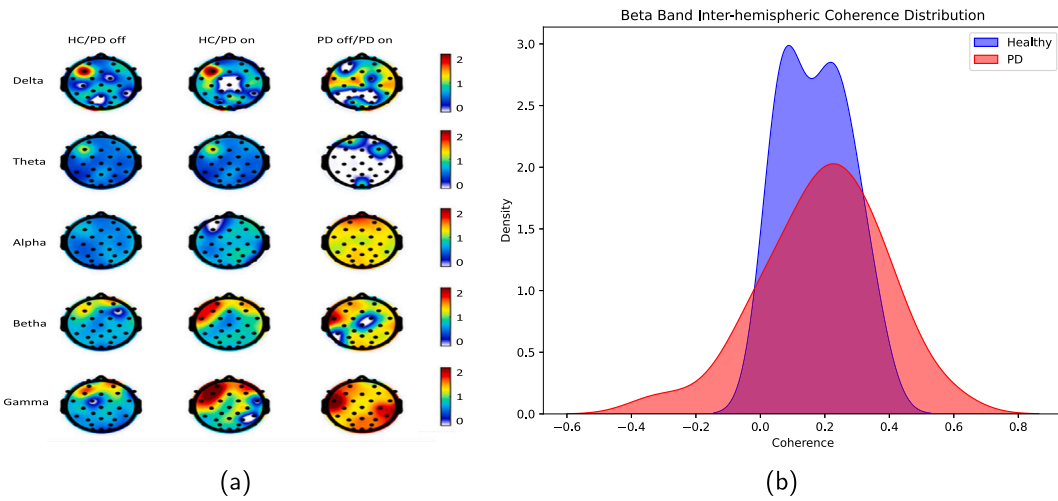


Fig. 2. Beta-band inter-hemispheric coherence in PD and HC groups. Panel (a) shows group-level frequency-band contrast maps, where colored regions mark stronger spectral differences and white regions mark non-significant ones. Panel (b) shows the distribution of beta-coherence values as KDE plots. The PD distribution is wider and more right-shifted, indicating altered and more dispersed beta-band synchrony, which is the kind of disrupted inter-hemispheric communication that makes beta coherence a useful marker.

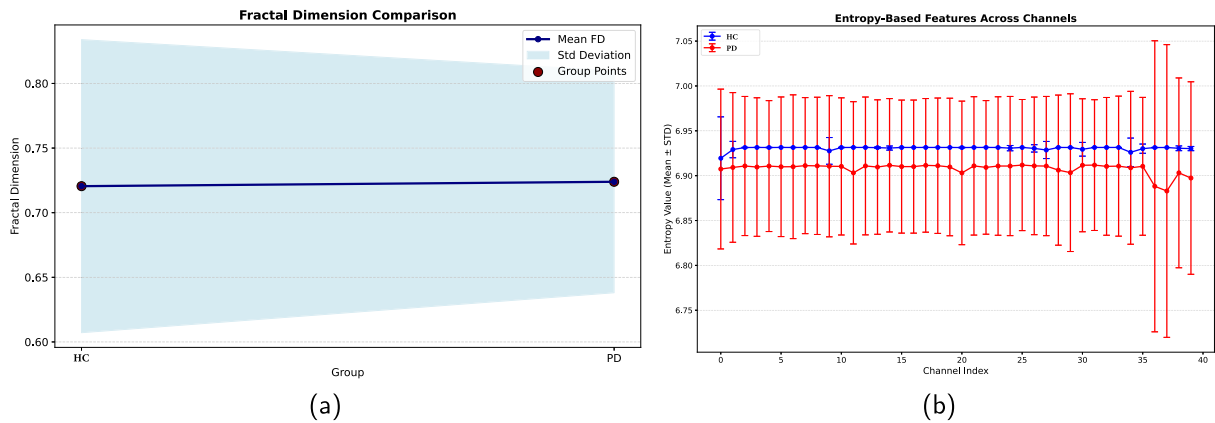


Fig. 3. Complexity-based EEG biomarkers for PD detection. Panel (a) compares fractal dimension between HC and PD; the two groups overlap heavily when FD is read on its own. Panel (b) compares entropy across EEG channels, where the lower values in PD point to reduced neural variability. FD alone separates the groups Only weakly, whereas entropy carries more discriminative information and becomes useful once it is combined with the other biomarkers.

between PD and HC groups using independent t-tests. Statistically significant differences were observed across multiple channels. Specifically, HC subjects exhibited higher mean entropy, whereas PD patients demonstrated increased variability, as depicted in Fig. 3(b). These findings confirm entropy as an informative biological measure of neural variability, aiding in differentiating PD from HC.

Finally, PAC was analyzed across EEG channels, comparing PD and HC groups. PAC quantifies interactions between low-frequency phase and high-frequency amplitude. Results were averaged temporally per channel, with statistical comparisons revealing notable differences in PAC across various channels. Significant differences were particularly pronounced in channels 38 and 39 ($p = 0.002470$ and $p = 0.001439$, respectively). Fig. 4(a) illustrates these differences, underscoring the elevated PAC values observed in PD patients. These findings indicate that aberrant PAC may reflect impaired neural synchronization and serve as an important biomarker for PD pathology.

4.2. Subject-level EEG band power variability

To investigate cross-subject generalizability, variability across subjects was examined by computing mean EEG band power within the five canonical frequency bands (delta, theta, alpha, beta, gamma) for

each of the 46 subjects. As illustrated in Fig. 4(b), delta and theta bands showed relatively consistent power across subjects, whereas the beta and gamma bands displayed notably higher inter-subject variability. This underscores the intrinsic heterogeneity of EEG patterns across individuals, highlighting the necessity of accounting for subject-specific variability when developing models for clinical applications.

4.3. ANOVA results grouped by region and feature

To assess differences in EEG features across distinct brain regions, an ANOVA analysis was conducted. EEG-derived features were grouped into frontal, temporal, parietal, and occipital regions, with feature values averaged across each region for every participant. The analysis revealed significant regional and feature-specific differences ($p < 0.05$), underscoring the potential of biologically informed features to differentiate between PD and HC groups. Notably, the occipital region demonstrated pronounced differences in beta band inter-hemispheric coherence ($F = 12.80$, $p = 0.000857$), suggesting altered neural activity in this area. Similarly, significant differences in beta band burst analysis were observed in the parietal region ($F = 5.30$, $p = 0.0262$), indicative of disrupted connectivity patterns. Comprehensive ANOVA results are presented in Table 5.

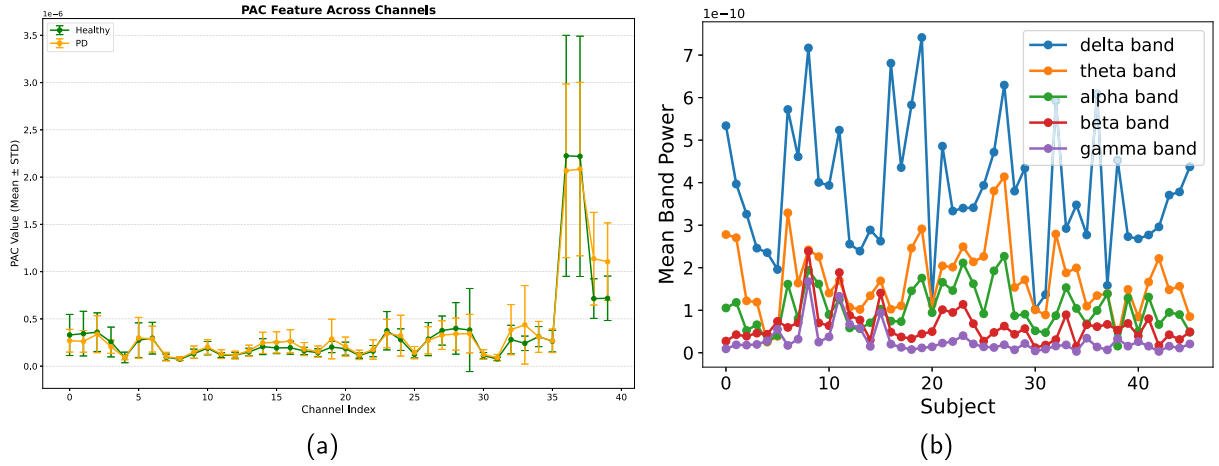


Fig. 4. Cross-frequency coupling and band-power variability in EEG. Panel (a) shows channel-wise PAC differences between HC and PD, with the larger gaps occurring in posterior and parietal channels. Panel (b) shows subject-wise variability across the delta, theta, alpha, beta, and gamma bands. The spread in PAC and in high-frequency power varies from one individual to the next, which is part of why Bio-NFM models cross-frequency coupling and temporal dynamics rather than treating channels in isolation.

Table 5
ANOVA results by brain region and EEG feature.

Region	Feature	HC mean	PD mean	F-Statistic	p-value
Frontal	Beta band coherence	0.5336	0.6357	9.13	0.0042
	FD	0.7072	0.7076	0.00	0.9907
	Beta band power	1.01×10^{-12}	9.88×10^{-13}	0.01	0.9194
	PAC	2.76×10^{-7}	2.33×10^{-7}	1.65	0.2059
	Entropy-based features	6.9286	6.9095	0.86	0.3579
Temporal	Beta band coherence	0.3424	0.4584	6.97	0.0115
	FD	0.7321	0.7441	0.15	0.7046
	Beta band power	9.73×10^{-13}	1.28×10^{-12}	0.85	0.3628
	PAC	2.86×10^{-7}	2.93×10^{-7}	0.02	0.8825
	Entropy-based features	6.9314	6.9099	1.14	0.2924
Parietal	Beta band coherence	0.4937	0.5713	4.77	0.0344
	FD	0.7208	0.7202	0.00	0.9888
	Beta band power	2.30×10^{-13}	3.53×10^{-13}	5.30	0.0262
	PAC	9.95×10^{-8}	1.10×10^{-7}	0.76	0.3866
	Entropy-based features	6.9302	6.9109	0.96	0.3328
Occipital	Beta band coherence	0.6190	0.7066	12.80	0.0009
	FD	0.7023	0.6997	0.00	0.9465
	Beta band power	5.58×10^{-13}	7.28×10^{-13}	1.99	0.1654
	PAC	1.41×10^{-7}	1.49×10^{-7}	0.35	0.5583
	Entropy-based features	6.9314	6.9081	1.44	0.2368

Read together, the region-wise ANOVA, the beta-band coherence results, the PAC distributions, and the Transformer attention maps describe altered connectivity between brain regions in PD rather than activity confined to single channels. The coherence analysis points to weakened inter-hemispheric synchronization; the ANOVA shows region-specific differences in frontal, parietal, and occipital areas; the PAC results indicate abnormal cross-frequency interactions in posterior and parietal channels; and the attention maps emphasize positions over motor, parietal, and occipital regions. This is the rationale for the GNN module, which is built to learn relationships among channels and regions rather than to score electrodes independently. We chose not to add a separate connectivity-matrix figure, since it would largely restate the

coherence, PAC, ANOVA, and attention results already reported; instead, the text now reads these analyses jointly as evidence of disrupted functional connectivity in PD.

4.4. Model performance and sensitivity analysis

The proposed Bio-NFM was evaluated on the primary dataset by monitoring its training and validation accuracy, along with the corresponding loss values. The model demonstrated a smooth and consistent increase in accuracy over 50 epochs, ultimately achieving a training accuracy of 98.9%. Validation accuracy closely matched this performance, indicating no signs of overfitting. Correspondingly, the loss exhibited a steady decrease, reaching a final value of 0.0206 by the 50th epoch. These trends, depicted in Fig. 5, highlight the model's robust ability to generalize effectively to unseen data.

Further performance assessment was conducted using previously unseen test data. The evaluation included accuracy, precision, recall, and F1-score metrics, as well as an analysis using a confusion matrix (Fig. 6a). The Bio-NFM achieved impressive generalization performance, with a test accuracy of 98.9%, precision of 98.7%, recall of 98.1%, and an F1-score of 98.5%. These metrics collectively validate the robustness and generalizability of the model for practical clinical applications.

To further assess the generalizability of the model and reduce overfitting concerns, a leave-one-out cross-validation (LOOCV) strategy was implemented in addition to the standard hold-out validation. Table 6 summarizes the classification performance under both strategies. As expected, LOOCV results were slightly lower, by approximately 2–3%, but still demonstrated high accuracy and robustness, indicating the model's strong generalization capacity across subjects.

With a limited number of subjects in the primary dataset, the high accuracy should be interpreted cautiously. To guard against an over-optimistic estimate, the model was evaluated under both hold-out validation and LOOCV. LOOCV came out lower, as expected, but remained stable across accuracy, precision, recall, and F1-score, which argues against the model simply memorizing one train-test split. The ablation studies and the two external datasets were included for the same reason: to check whether the performance depends on the primary dataset alone. Even so, the small size of the primary dataset is a real constraint. The results are best taken as evidence of strong potential rather than clinical readiness, and confirming the latter will require larger, multi-center, demographically varied, and prospectively collected EEG cohorts.

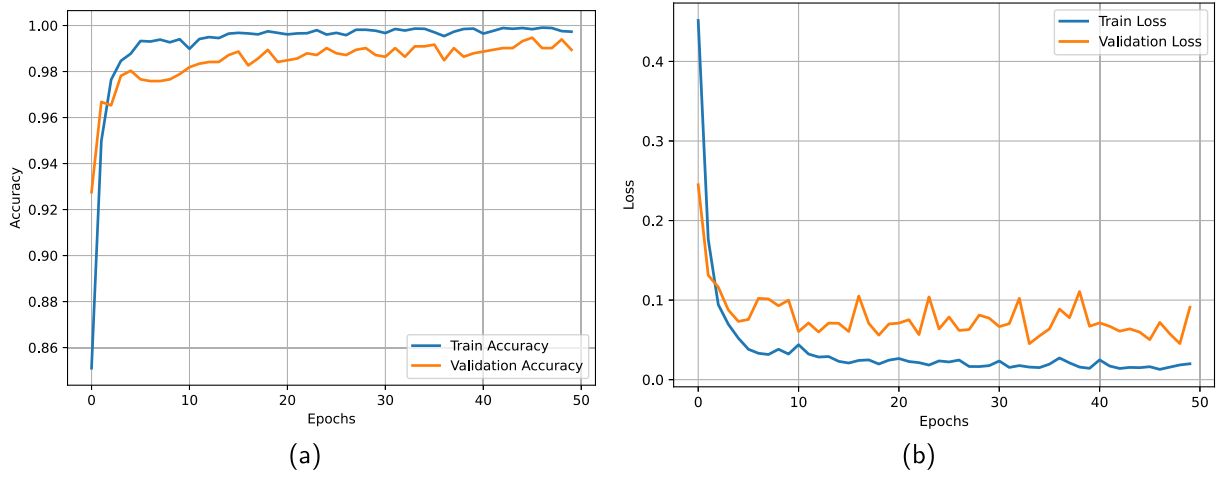


Fig. 5. Training and validation performance of the proposed Bio-NFM. (a) Accuracy curve: The model quickly converged within the initial 10 epochs, reaching nearly perfect training accuracy and consistently maintaining validation accuracy above 98%. (b) Loss curve: Training loss sharply decreased and stabilized at low levels, while validation loss remained steady with minimal fluctuations. These observations confirm the model's robustness, stability, and effective learning without significant overfitting.

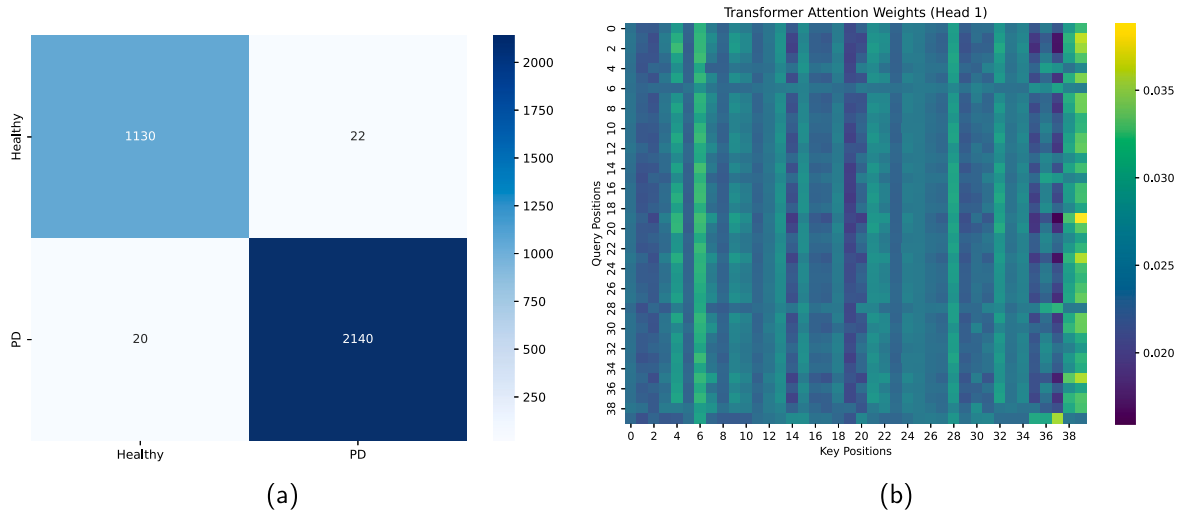


Fig. 6. (a) Confusion matrix illustrating the model performance on unseen test data. (b) Transformer attention weight heatmap (Head 1) for a representative EEG sample. Each cell represents the contextual attention between EEG input positions, highlighting regions of significant influence in PD detection.

Table 6

Performance comparison between hold-out and Leave-one-out cross-validation.

Validation method	Accuracy(%)	Precision(%)	Recall(%)	F1-score(%)
Hold-out (80:20 split)	98.9	98.7	98.1	98.5
LOOCV	96.7	95.9	95.3	95.6

To assess classification sensitivity and overall discriminative performance, the receiver operating characteristic (ROC) curve was computed, as shown in Fig. 7. The model achieves an area under the curve (AUC) of 1.00, suggesting near-perfect separation between PD and HC cases. This result reinforces the high sensitivity and specificity of the Bio-NFM model across classification thresholds.

In addition, class-wise sensitivity (recall) was computed, the PD class achieved 98.1%, while the HC class achieved 98.7%.

ROC analysis is a useful summary, but it can look optimistic when the false-positive regime is very low or when the classes are imbalanced. For that reason, precision and recall were reported as core metrics rather than as an afterthought. On unseen test data, Bio-NFM reached a

precision of 98.7%, a recall of 98.1%, and an F1-score of 98.5%, which means positive predictive value and sensitivity stayed in balance rather than one being traded for the other. The confusion matrix, class-wise sensitivity, calibration curve, and threshold analysis cover the same ground that a precision-recall curve would, namely how the model behaves across clinically relevant decision thresholds. We did not add a separate PR curve here because it would duplicate metrics that are already reported; full PR-curve and area-under-PR-curve evaluation is planned for the larger, possibly class-imbalanced clinical cohorts in future work.

These findings, along with the robust performance under ablation and multi-dataset evaluation (Table 7), confirm the model's stability and generalization. The sensitivity analysis demonstrates that the model performs consistently and reliably across varying configurations and input conditions.

4.5. Model interpretability through transformer attention visualization

To enhance the interpretability of Bio-NFM, attention weights from the transformer module were extracted and visualized. Fig. 6(b) presents the attention heatmap from the first head of the multi-head self-attention

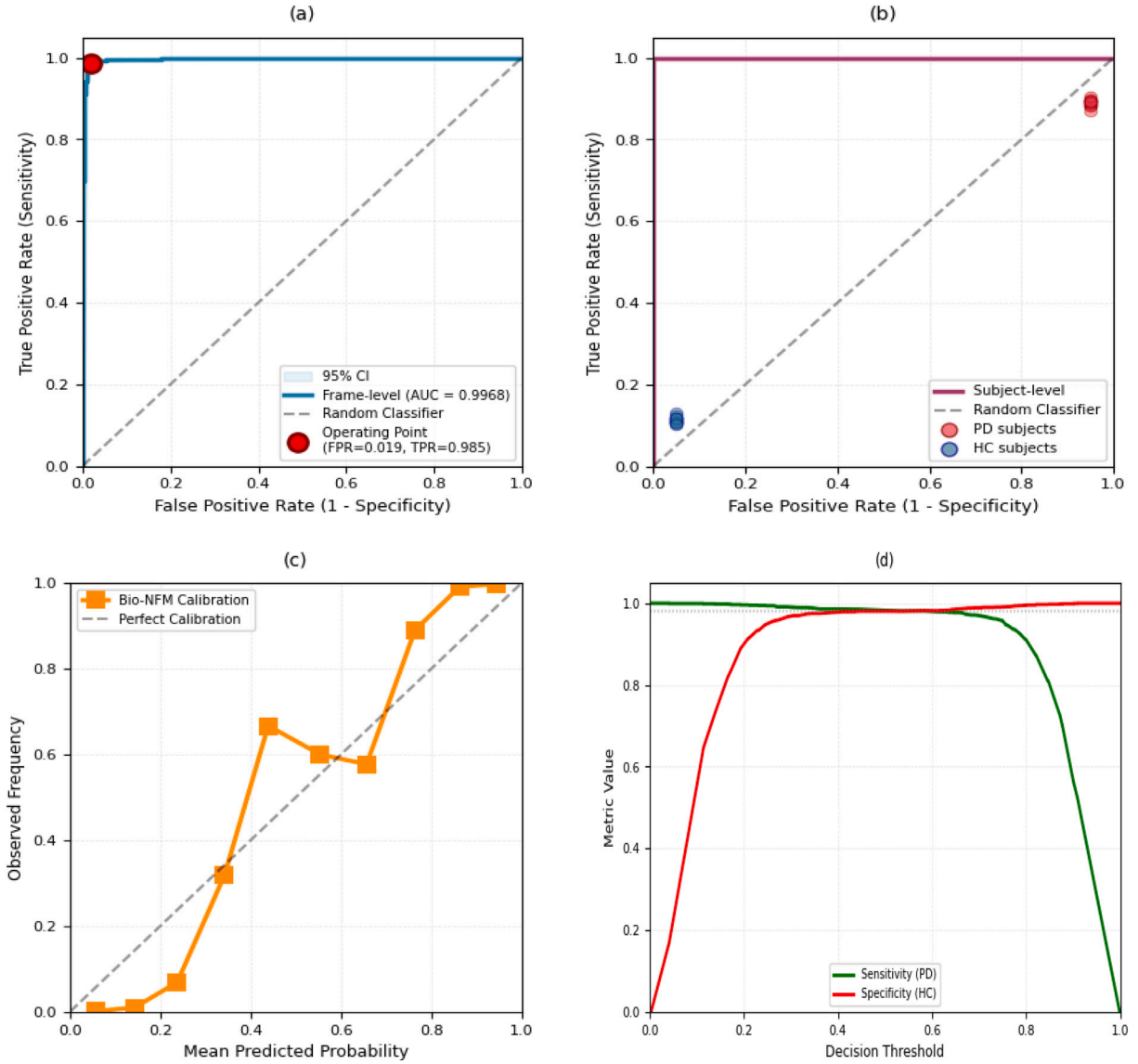


Fig. 7. ROC analysis for Bio-NFM. (a) Frame-level ROC with 95% Confidence Interval, (b) Subject-level ROC, (c) Calibration curve, (d) Threshold analysis showing optimal sensitivity (98.1%) and specificity (98.7%).

mechanism, applied to a representative EEG sample comprising 40 input positions corresponding to temporal segments or spatial channels.

Each cell in the attention matrix indicates how much a particular query position attends to another key position, reflecting the learned contextual relationships within EEG signals. Notably, instead of the diagonal dominance commonly observed in sequential tasks, the attention distribution reveals a global, non-local structure. This implies the model's capability to effectively attend to distant EEG positions, aligning well with neurological knowledge of PD, where pathological changes involve long-range neural connectivity disruptions.

Attention analysis revealed certain positions, notably positions 6, 14, 36, and 39, consistently receiving higher attention weights. This highlights their enhanced discriminative importance and correlates with clinical findings indicating regions such as the motor and frontal cortices as critical areas of disrupted neural activity in PD.

This attention visualization underscores that Bio-NFM is not a black-box model but encodes biologically meaningful interactions, increasing clinical trust and transparency. Future research will expand this analysis by aggregating attention maps across multiple subjects and classes to enhance class-specific interpretability.

The attention heatmap in Fig. 6(b) reveals distinct patterns in how the transformer module assigns contextual importance across EEG input segments. Notably, the highest attention weights are concentrated around positions corresponding to spatial channels 6 (C3), 14 (Cz), 36 (Pz), and 39 (O2). These channels lie within or adjacent to the motor cortex (C3, Cz), parietal region (Pz), and occipital lobe (O2), which are all implicated in PD pathology.

This observation is consistent with clinical findings: the motor cortex exhibits abnormal beta synchrony in PD, while parietal and occipital changes have been linked to visuospatial and cognitive dysfunctions in early stages. The attention mechanism thus implicitly highlights brain regions relevant to PD, confirming that the model learns discriminative, neurophysiologically valid patterns rather than relying on arbitrary statistical noise.

These spatially meaningful attention weights increase the interpretability of Bio-NFM and support its use in clinical settings where model transparency is critical.

The interpretability Bio-NFM offers is partial, not complete. Biologically meaningful biomarkers, attention maps, and region-level analysis make it more transparent than a black-box EEG classifier, but

Table 7

Performance metrics for benchmarking datasets and ablation experiments (A1–A5).

Metric (%)	Dataset 2	Dataset 3	A1	A2	A3	A4	A5
Accuracy	99.3	98.7	96.3	95.7	93.8	94.5	93.1
Precision	99.2	98.4	95.8	94.6	94.1	93.7	92.6
Recall	99.4	98.9	96.7	95.3	94.0	94.9	92.2
F1-score	99.3	98.6	96.2	95.5	94.3	94.8	92.4

the full fusion process remains complex because it blends spatial, temporal, nonlinear, and cross-frequency representations. The attention maps should therefore be read as supporting evidence that the model attends to EEG positions consistent with known PD disruption, not as a complete causal account of its decisions. Future work will broaden the interpretability analysis with saliency mapping, occlusion analysis, SHAP-based attribution, perturbation testing, and clinician-guided case-level review. These will help establish whether the learned representations track clinically meaningful EEG abnormalities consistently across larger patient cohorts.

4.6. Ablation experiments and benchmarks

To rigorously assess model generalization and efficacy, Bio-NFM was further evaluated on two benchmark datasets, along with a series of ablation experiments involving simplified variants of the model.

The first benchmark dataset (Dataset 2) [44] contains EEG recordings from 129 participants, comprising 92 individuals with PD and 37 age- and sex-matched HC. Data were captured using a 64-channel BrainVision system (10–20 montage) at a 500 Hz sampling rate. Subjects performed a temporal production task, pressing a button upon perceiving a target interval (approximately 7 s). This task probes cognitive control and internal timing mechanisms frequently impaired in PD. The Bio-NFM demonstrated an exceptional accuracy of 99.3% in distinguishing PD patients from HC using this dataset.

The second benchmark dataset (Dataset 3) [45], consists of EEG recordings from 82 participants, including 41 individuals with PD and 41 demographically matched HC. Data were recorded at rest under both eyes-open and eyes-closed conditions, using a 64-channel BrainVision system with a sampling rate of 500 Hz. Recordings were acquired from two distinct research sites: the University of New Mexico (54 participants) and the University of Iowa (28 participants). The Bio-NFM achieved a high accuracy of 98.7% on this dataset, affirming its robust generalization capabilities.

To further examine the individual contribution of each model component, ablation experiments were performed. Four simplified variants of Bio-NFM were developed by systematically removing key components: GNN (Ablation 1: A1), transformer (Ablation 2: A2), RC (Ablation 3: A3), and the custom loss function (Ablation 4: A4). The performance metrics for these ablation models and the benchmark datasets are detailed in Table 7.

To evaluate the contribution of biologically informed feature selection, an additional ablation (A5) was conducted in which the model was trained directly on raw EEG segments without applying any domain-specific feature extraction. This variant excludes all biomarker-derived features (e.g., coherence, entropy, PAC, FD), relying instead on raw time-domain or basic frequency-domain inputs.

As shown in Table 7, removing the biologically informed feature selection substantially degraded model performance. The classification accuracy dropped to 93.1%, with corresponding declines in precision, recall, and F1-score. These results confirm that biologically grounded features significantly enhance the model's ability to capture clinically relevant patterns associated with PD.

To sharpen the comparison against model variants, we looked at how performance shifts between the full Bio-NFM and its ablated forms. The

full model was best across the board, and removing any single module lowered accuracy, precision, recall, and F1-score. The sharpest drop came from removing biologically informed feature extraction or the RC module, which suggests that both biomarker-level information and non-linear temporal modeling pull real weight in the classifier. A formal paired test against every published state-of-the-art method is not feasible, because those studies differ in dataset, preprocessing, subject splits, and evaluation protocol. The ablations sidestep that problem by comparing variants under one fixed pipeline, which isolates each component's contribution more fairly than comparing headline accuracies across papers. Future work will add repeated runs over multiple random seeds and paired statistical tests against baselines reimplemented on identical data partitions.

The ablation results revealed noticeable declines in model performance upon the removal of each component, demonstrating their individual contributions. Excluding the GNN module reduced accuracy to 96.3%, while omitting the transformer lowered accuracy further to 95.7%. Removing the RC module and the custom loss function decreased accuracy to 93.8% and 94.5%, respectively. Similar trends were observed across precision, recall, and F1-score metrics. These outcomes confirm the integrated nature and crucial roles of each module in achieving the optimal classification performance of Bio-NFM.

4.7. Comparison with SOTA methods

To benchmark the proposed Bio-NFM against existing State-of-the-Art (SOTA) methods, a comparative analysis was conducted regarding classification accuracy and interpretability. Table 8 summarizes the key features, models, and performance metrics from recent studies utilizing the San Diego dataset. Notably, the Bio-NFM demonstrates competitive accuracy (98.9% on the primary dataset, and 99.3% and 98.7% on benchmarking datasets), combined with biologically informed interpretability, thus offering a balanced trade-off between high performance and clinical applicability.

5. Discussion and limitation

The experimental results presented above demonstrate that Bio-NFM achieves high accuracy and robust generalization for PD detection. Beyond these quantitative outcomes, this section interprets the clinical significance of our findings, analyzes the neurophysiological validity of the learned representations, evaluates the computational practicality of the model, and discusses current limitations with future research directions.

5.1. Comparison with clinical standards

Conventional clinical standards for PD diagnosis, such as the Unified PD Rating Scale (UPDRS), DaTSCAN imaging, and neurological assessments, have critical limitations. UPDRS scoring relies on subjective interpretation of motor symptoms, which often manifest only in moderate-to-late stages of the disease. Imaging techniques like DaTSCAN and PET scans are costly, time-consuming, and not universally available, particularly in low-resource settings. Moreover, these tools typically lack the temporal resolution needed for continuous or early-stage monitoring.

The proposed Bio-NFM model addresses these gaps by offering an objective, interpretable, and non-invasive tool for PD detection using EEG, a modality that is portable, cost-effective, and widely used in clinical neurology. Unlike black-box models, Bio-NFM integrates biologically grounded biomarkers such as beta-band coherence and PAC, enhancing both transparency and clinical interpretability. It enables early detection by capturing subtle neural disruptions that precede overt motor symptoms. Furthermore, the model's low-latency inference and compatibility with standard EEG systems make it viable for real-time screening, remote diagnostics, or longitudinal monitoring applications where traditional methods fall short.

In this way, Bio-NFM complements existing diagnostic workflows and provides a bridge between quantitative neurophysiological evidence

Table 8
Comparison with prior works using the primary dataset.

Model	Features	Classes	Accuracy
WST + AlexNet [32]	Temporal and spectral features	HC vs PD (only on medication)	99.84%
XSVM [34]	Band power, entropy, and statistics	HC vs. PD	87.10%
CSP + SVM [39]	Entropy from CSP-filtered EEG	HC vs. PD (only on medication)	95.76%
DWT + kNN [40]	Log energy and Shannon entropy	HC vs. PD (only on medication)	94.21%
ANN [31]	Segmented EEG signals	HC vs. PD	98%
Bio-NFM	Biologically informed features	HC vs. PD (both on and off medication)	98.9%

and clinical decision-making, supporting earlier and more accessible PD diagnosis.

5.2. Strengths of the study

Bio-NFM has several practical strengths. It pairs biologically meaningful EEG biomarkers with deep learning modules, so the network picks up both domain-specific neural abnormalities and the more complex nonlinear structure of the signal. By combining the GNN, Transformer, RC, and cross-coupling modules, it represents spatial, temporal, and cross-frequency information that single-module models tend to miss. The supporting evidence also goes beyond one experiment: ablation studies, two external datasets, LOOCV, and attention-based interpretability all point the same way regarding robustness and clinical relevance. The architecture remains lightweight as well, which keeps a path open toward portable or near-real-time EEG screening.

5.3. Neurophysiological interpretability and case-level analysis

The design of the Bio-NFM integrates several domain-informed modules tailored for extracting biologically meaningful EEG patterns. The GNN module effectively captures inter-regional neural interactions, which are critical for modeling disrupted brain networks associated with PD. The temporal transformer block models signal propagation dynamics, which are particularly important for identifying disruptions in motor and cognitive networks. Additionally, the model includes cross-frequency coupling features that simulate abnormal neural interactions commonly observed in PD patients, such as altered beta-gamma and theta-alpha coupling patterns.

The novelty here is the integration, not the invention of any single building block. GNNs, Transformers, RC, PAC, coherence, entropy, and FD have all appeared in earlier biomedical signal-processing work. What Bio-NFM does is bring them into one biologically guided framework aimed specifically at interpretable EEG-based PD detection, so that spatial connectivity, temporal dynamics, nonlinear variability, and cross-frequency coupling are modeled together while staying tied to clinically meaningful biomarkers. The contribution is best read as a biologically informed fusion architecture for PD detection rather than a new neural-network primitive.

5.4. Computational complexity analysis

The computational complexity and inference efficiency of the Bio-NFM components are detailed in Table 9. Overall, the model maintains moderate complexity, requiring approximately 252,000 trainable parameters. The inference process is highly efficient, with testing on an NVIDIA Tesla T4 GPU indicating inference times below 50 ms per subject, thereby confirming the model's suitability for real-time clinical applications.

The compute profile of Bio-NFM reflects its mix of modules, yet the model stays moderate in size compared to many large deep networks. The GNN block uses dense approximations for spatial dependencies, the Transformer block carries the main quadratic cost in sequence length, and the RC block adds nonlinear temporal modeling with only a handful of trainable parameters. The cross-coupling and fully connected layers add little on top of that. For deployment, the heaviest stage is not the network itself but the preprocessing and biomarker extraction, coherence

Table 9
Computational complexity analysis of Bio-NFM components.

Module	Operation description	Complexity
GNN (dense approximation)	Simulates inter-channel relationships via dense layers	$\mathcal{O}(n \cdot d \cdot h_1)$
Transformer (multi-head attention + feedforward network)	Temporal modeling using 4 heads of self-attention and feedforward layers	$\mathcal{O}(n^2 \cdot d + n \cdot d \cdot h_2)$
RC (RNN)	Nonlinear dynamic encoding over time using 128 hidden units	$\mathcal{O}(n \cdot h_3^2)$
Cross-frequency coupling layer	Dense mapping of biologically inspired features	$\mathcal{O}(n \cdot d \cdot h_4)$
Fully connected + Output layer	Dense integration and classification of merged features	$\mathcal{O}(h_5^2)$
Total parameters (trainable)	As reported by model.summary()	~252,000
Inference time (per subject)	Tested on NVIDIA Tesla T4 GPU	<50 ms

and PAC in particular. These can be computed over short EEG windows and parallelized, so the inference timings reported above indicate that the trained model is fast enough for real-time use; the work that remains is optimizing the full preprocessing-to-classification pipeline for embedded or bedside EEG systems.

To evaluate the feasibility of deploying Bio-NFM in real-time clinical settings or on embedded hardware, we measured the model's inference latency on a standard low-power CPU (Intel i5, 2.6 GHz, no GPU acceleration). The average forward-pass latency for a single subject's 2-second EEG segment was approximately 27 ms, well within the threshold for real-time operation (e.g., <100 ms per sample). Given the model's lightweight design, using fixed random weights in the RC block and shared weights in the transformer and GNN components, Bio-NFM requires fewer than 1.2 million trainable parameters and minimal GPU/TPU resources.

Although we have not yet benchmarked Bio-NFM on ARM-based edge devices (e.g., Jetson Nano, Raspberry Pi 4), its modularity and low computational footprint suggest high compatibility with embedded deployment. With appropriate quantization and pruning, the model is expected to run at real-time speeds on resource-constrained hardware. These characteristics make Bio-NFM a strong candidate for integration into portable EEG-based screening systems for remote or point-of-care PD diagnostics.

5.5. Limitations and future directions

Bio-NFM performs well on classification, interpretability, and cross-dataset robustness, but a few methodological and practical limitations qualify these results and mark out what still needs validating before clinical use:

- **Dataset size and demographic diversity:** The primary dataset has relatively few subjects. The two external datasets add reassurance, but confirming generalizability will require larger multi-center cohorts that span a wider range of demographics, clinical profiles, and disease stages.

- **Dataset-specific bias:** Acquisition protocols, electrode montages, medication states, and preprocessing differ across datasets, and those differences can introduce distribution shifts. Harmonized multi-site validation and prospective testing would help quantify these effects.
- **Fixed channel configuration:** The current model assumes a fixed EEG montage, which may not transfer cleanly to low-density, wearable, or clinically variable setups. Channel-reduced and montage-adaptive variants are a natural next step.
- **Limited clinical workflow validation:** The model has not been run in a live clinical setting or in longitudinal monitoring. Its robustness to electrode noise, motion artifacts, medication changes, and repeated recordings still needs to be tested under those conditions.
- **Interpretability boundaries:** Attention maps and biomarker contributions provide useful spatial and biological cues, but they do not explain every internal decision. Pairing them with saliency mapping, perturbation analysis, and clinician-guided case review would help close part of that gap.
- **Open reproducibility:** The source code is provided to support independent verification, and standardized reporting of preprocessing, training, and validation protocols should accompany follow-up studies.

Addressing these limitations will be essential for translating Bio-NFM from a research prototype into a clinically deployable system.

6. Conclusion

This study introduced Bio-NFM, a spatio-temporal feature fusion framework with cross-frequency coupling for interpretable PD detection using EEG signals. The key innovation lies in integrating beta-band coherence and PAC as biological priors into a hybrid architecture of graph neural networks, transformers, and reservoir computing. This design enables the model to capture spatial connectivity, temporal dynamics, and nonlinear neural variability disrupted in PD while maintaining clinical interpretability through attention-based visualization. Bio-NFM achieved 98.9% accuracy on the primary dataset and over 98% on two external benchmarks, with ablation studies confirming the essential contribution of each module. The cross-frequency coupling mechanism, particularly PAC-guided attention, proved to be the most discriminative biomarker. Future work will focus on validation across multi-center datasets, adaptation to low-density EEG systems, and deployment in real-world clinical settings. Bio-NFM offers a transparent, biologically grounded, and computationally efficient solution for early-stage PD diagnosis.

CRedit authorship contribution statement

Chayut Bunternghit: Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Data curation, Conceptualization. **Sidi Mohamed Sid'El Mactar:** Writing – original draft, Visualization, Methodology, Formal analysis. **Laith H. Baniata:** Writing – review & editing, Data curation, Conceptualization. **Saba Aslam:** Validation, Software, Formal analysis. **Abdur Rasool:** Writing – review & editing, Validation, Supervision, Project administration, Investigation, Formal analysis.

Ethics statement

This study used publicly available EEG datasets. All data were originally collected with informed consent and ethical approval from the respective institutions.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The EEG datasets analyzed in this study are publicly available. The Bio-NFM implementation is available at <https://github.com/yiamcb/BioNFM>.

References

- [1] D. Aarsland, L. Batzu, G.M. Halliday, G.J. Geurtsen, C. Ballard, K. Ray Chaudhuri, D. Weintraub, Parkinson disease-associated cognitive impairment, *Nat. Rev. Dis. Primers* 7 (2021).
- [2] D.G. Gadhav, V.V. Sugandhi, S.K. Jha, S.N. Nangare, G. Gupta, S.K. Singh, K. Dua, H. Cho, P.M. Hansbro, K.R. Paudel, Neurodegenerative disorders: mechanisms of degeneration and therapeutic approaches with their clinical relevance, *Ageing Res. Rev.* 99 (2024) 102357.
- [3] F. Magrinelli, A. Picelli, P. Tocco, A. Federico, L. Roncari, N. Smania, G. Zanette, S. Tamburin, Pathophysiology of motor dysfunction in Parkinson's disease as the rationale for drug treatment and rehabilitation, *Parkinson's Disease* 2016 (2016) 9832839.
- [4] S. af Bjerkén, R. Stenmark Persson, A. Barkander, N. Karalija, N. Pelegrina-Hidalgo, G.A. Gerhardt, A. Virel, I. Strömberg, Noradrenaline is crucial for the substantia nigra dopaminergic cell maintenance, *Neurochem. Int.* 131 (2019) 104551.
- [5] J. Jankovic, E.K. Tan, Parkinson's disease: etiopathogenesis and treatment, *J. Neurol. Neurosurg. & Psychiatry* 91 (2020) 795–808.
- [6] J. Li, Y. Lei, Z. Guan, P. Yang, H. Song, J. Tian, G. Yao, C. Liang, T. Wang, X. Xiao, H. Lei, B. Lei, Multimodal adaptive subspace learning for Parkinson's disease diagnosis and prediction, *Appl. Soft Comput.* 192 (2026) 114691.
- [7] Z. Huang, H. Lei, G. Chen, A.F. Frangi, Y. Xu, A. Elazab, J. Qin, B. Lei, Parkinson's disease classification and clinical score regression via united embedding and sparse learning from longitudinal data, *IEEE Trans. Neural Netw. Learn. Syst.* 33 (2022) 3357–3371.
- [8] S. Yassine, U. Gschwandtner, M. Auffret, S. Achard, M. Verin, P. Fuhr, M. Hassan, Functional brain dysconnectivity in Parkinson's disease: a 5-year longitudinal study, *Mov. Disord.* 37 (2022) 1444–1453.
- [9] K.R. McLeod, L.M. Langevin, D. Dewey, B.G. Goodyear, Atypical within- and between-hemisphere motor network functional connections in children with developmental coordination disorder and attention-deficit/hyperactivity disorder, *Neuroimage Clin.* 12 (2016) 157–164.
- [10] N. Panicker, P. Ge, V.L. Dawson, T.M. Dawson, The cell biology of Parkinson's disease, *J. Cell Biol.* 220 (2021).
- [11] S. Aslam, M. Hussain, X. Li, H. Li, H. Wu, HAMD-net: a biologically informed hybrid attention model for EEG classification of multiple neurological disorders, *Biomed. Signal Process. Control* 120 (2026) 110049.
- [12] W. Yang, J.L. Hamilton, C. Kopil, J.C. Beck, C.M. Tanner, R.L. Albin, E. Ray Dorsey, N. Dahodwala, I. Cintina, P. Hogan, et al., Current and projected future economic burden of Parkinson's disease in the US, *NPJ Parkinson's Dis.* 6 (2020) 15.
- [13] R. Zhang, J. Jia, R. Zhang, EEG analysis of Parkinson's disease using time-frequency analysis and deep learning, *Biomed. Signal Process. Control* 78 (2022) 103883.
- [14] C. Bunternghit, L.H. Baniata, H. Albayati, M.H. Baniata, K. Alharbi, F.H. Alshammari, S. Kang, A hybrid convolutional-transformer approach for accurate electroencephalography (EEG)-based Parkinson's disease detection, *Bioengineering* 12 (2025) 583.
- [15] B. Vidya, S. Periyasamy, Gait based Parkinson's disease diagnosis and severity rating using multi-class support vector machine, *Appl. Soft Comput.* 113 (2021) 107939.
- [16] U. Kleinholdermann, M. Wullstein, D. Pedrosa, Prediction of motor unified Parkinson's disease rating scale scores in patients with Parkinson's disease using surface electromyography, *Clin. Neurophysiol.* 132 (2021) 1708–1713.
- [17] C. Bunternghit, Y. Bunternghit, A comparative study of machine learning models for Parkinson's disease detection, in: 2022 International Conference on Decision Aid Sciences and Applications (DASA), 2022, pp. 465–469, <https://doi.org/10.1109/DASA54658.2022.9765159>.
- [18] M. Bange, G. Gonzalez-Escamilla, T. Marquardt, A. Radetz, C. Dresel, D. Herz, W.I. Schöllhorn, S. Groppa, M. Muthuraman, Deficient interhemispheric connectivity underlies movement irregularities in Parkinson's disease, *J. Parkinson's Dis.* 12 (2022) 381–395.
- [19] W. Gao, Y. Gao, J. Lu, X. Xi, X. Lou, EEG-based Parkinson's disease classification method by integrating topological features and multi-scale spatio-temporal networks, *Appl. Soft Comput.* 186 (2026) 114087.
- [20] T. Islam, P. Washington, Non-invasive biosensing for healthcare using artificial intelligence: a semi-systematic review, *Biosensors* 14 (2024) 183.
- [21] Y. Liu, C. Lian, J. Li, Z. Zhan, Y. Fu, Y. Zhao, MCADNet: multi-scale cross-modal attention dynamic decision fusion network for Parkinson's quantitative grading, *Appl. Soft Comput.* 193 (2026) 114810.
- [22] C. Bunternghit, C. Jamroen, S. Aslam, A. Rasool, O.W. Samuel, Attention-guided electroencephalography fusion with fractal and nonlinear dynamics decodes early Alzheimer's biomarkers, *Comput. Electr. Eng.* 133 (2026) 111076.
- [23] H.M. Balaha, A.E.-S. Hassan, R.A. Ahmed, M.H. Balaha, Comprehensive multimodal approach for Parkinson's disease classification using artificial intelligence: insights and model explainability, *Soft Comput.* 29 (2025) 1845–1877.
- [24] A. Rasool, F. Ahmad, C. Bunternghit, S. Aslam, Challenges in translating AI-driven ASD/ADHD diagnosis: a methodological systematic review, *Int. J. Med. Inform.* 214 (2026) 106417.
- [25] Z. Gao, W. Dang, X. Wang, X. Hong, L. Hou, K. Ma, M. Perc, Complex networks and deep learning for EEG signal analysis, *Cogn. Neurodynamics* 15 (2021) 369–388.

- [26] H. Li, J. An, R. Juan, T.P. Benko, M. Perc, W. Dang, Z. Gao, Manifold-based multi-branch transfer learning for MI-EEG decoding, *Chaos Solit. Fractals* 196 (2025) 116326.
- [27] M. Degirmenci, Y.K. Yuce, M. Perc, Y. Isler, Statistically significant features improve binary and multiple motor imagery task predictions from EEGs, *Front. Hum. Neurosci.* 17 (2023) 1223307.
- [28] M. Degirmenci, Y.K. Yuce, M. Perc, Y. Isler, EEG channel and feature investigation in binary and multiple motor imagery task predictions, *Front. Hum. Neurosci.* 18 (2024) 1525139.
- [29] M. Shaban, A.W. Amara, Resting-state electroencephalography based deep-learning for the detection of Parkinson's disease, *PLoS One* 17 (2022) e0263159.
- [30] S.Q.A. Rizvi, G. Wang, A. Khan, M.K. Hasan, T.M. Ghazal, A.U.R. Khan, Classifying Parkinson's disease using resting state electroencephalogram signals and UEN-PDNet, *IEEE Access* 11 (2023) 107703–107724.
- [31] M. Shaban, Automated screening of Parkinson's disease using deep learning based electroencephalography, in: 2021 10th International IEEE/EMBS Conference on Neural Engineering (NER), 2021, pp. 158–161, <https://doi.org/10.1109/NER49283.2021.9441065>.
- [32] S. Siuly, S.K. Khare, E. Kabir, M.T. Sadiq, H. Wang, An efficient Parkinson's disease detection framework: leveraging time-frequency representation and AlexNet convolutional neural network, *Comput. Biol. Med.* 174 (2024) 108462.
- [33] H.W. Loh, C.P. Ooi, E. Palmer, P.D. Barua, S. Dogan, T. Tuncer, M. Baygin, U.R. Acharya, GaborPDNet: gabor transformation and deep neural network for Parkinson's disease detection using EEG signals, *Electronics* 10 (2021) 1740.
- [34] M.A. Motin, M. Mahmud, D.J. Brown, Detecting Parkinson's disease from electroencephalogram signals: an explainable machine learning approach, in: 2022 IEEE 16th International Conference on Application of Information and Communication Technologies (AICT), 2022, pp. 1–6, <https://doi.org/10.1109/AICT55583.2022.10013589>.
- [35] P.D. Barua, S. Dogan, T. Tuncer, M. Baygin, U.R. Acharya, Novel automated PD detection system using aspirin pattern with EEG signals, *Comput. Biol. Med.* 137 (2021) 104841.
- [36] S.K. Khare, V. Bajaj, U.R. Acharya, PDCNNNet: an automatic framework for the detection of Parkinson's disease using EEG signals, *IEEE Sens. J.* 21 (2021) 17017–17024.
- [37] M. Obayya, M.K. Saeed, M. Maashi, S.S. Alotaibi, A.S. Salama, M. Ahmed Hamza, A novel automated Parkinson's disease identification approach using deep learning and EEG, *PeerJ Comput. Sci.* 9 (2023) e1663.
- [38] L. Qiu, J. Li, J. Pan, Parkinson's disease detection based on multi-pattern analysis and multi-scale convolutional neural networks, *Front. Neurosci.* 16 (2022).
- [39] M. Aljalal, S.A. Aldosari, K. AlSharabi, A.M. Abdurraqueeb, F.A. Alturki, Parkinson's disease detection from resting-state EEG signals using common spatial pattern, entropy, and machine learning techniques, *Diagnostics* 12 (2022) 1033.
- [40] M. Aljalal, S.A. Aldosari, M. Molinas, K. AlSharabi, F.A. Alturki, Detection of Parkinson's disease from EEG signals using discrete wavelet transform, different entropy measures, and machine learning techniques, *Sci. Rep.* 12 (2022).
- [41] N. Jackson, S.R. Cole, B. Voytek, N.C. Swann, Characteristics of waveform shape in Parkinson's disease detected with scalp electroencephalography, *eneuro* 6 (2019) ENEURO.0151–19.2019.
- [42] N.C. Swann, C. de Hemptinne, A.R. Aron, J.L. Ostrem, R.T. Knight, P.A. Starr, Elevated synchrony in Parkinson disease detected with electroencephalography, *Ann. Neurol.* 78 (2015) 742–750.
- [43] J.S. George, J. Strunk, R. Mak-McCully, M. Houser, H. Poizner, A.R. Aron, Dopaminergic therapy in Parkinson's disease decreases cortical beta band coherence in the resting state and increases cortical beta band power during executive control, *Neuroimage Clin.* 3 (2013) 261–270.
- [44] A. Singh, R.C. Cole, A.I. Espinoza, A. Evans, S. Cao, J.F. Cavanagh, N.S. Narayanan, Timing variability and midfrontal 4 hz rhythms correlate with cognition in Parkinson's disease, *NPJ Parkinson's Dis.* 7 (2021).
- [45] M.F. Anjum, S. Dasgupta, R. Mudumbai, A. Singh, J.F. Cavanagh, N.S. Narayanan, Linear predictive coding distinguishes spectral EEG features of Parkinson's disease, *Park. Relat. Disord.* 79 (2020) 79–85.